

Vectors and Coordinates #3

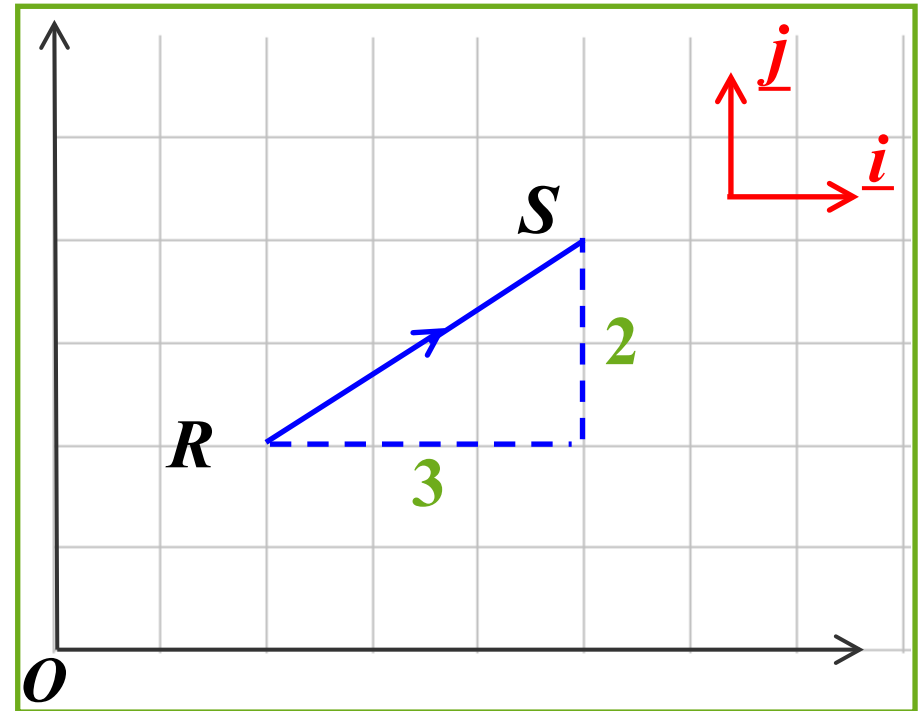


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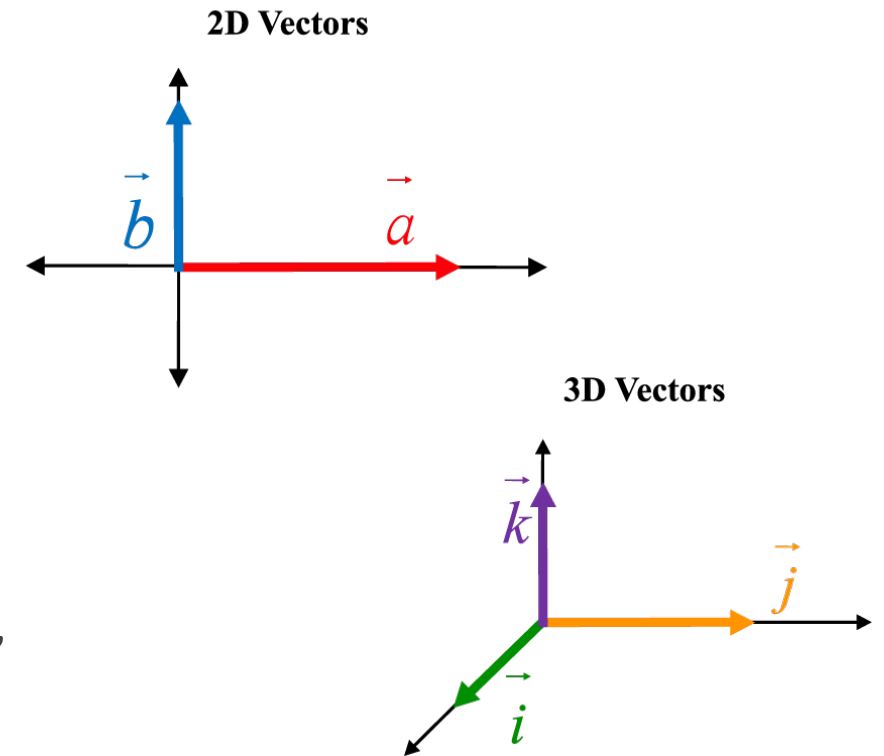
i and j

- If we don't want to constantly draw out our vectors or clarify them with $\overrightarrow{AB}$ we can write out a vector with i and j
- i is any movement in the x axis
- j is any movement in the y axis
- So we can write $v = 3i + 2j$



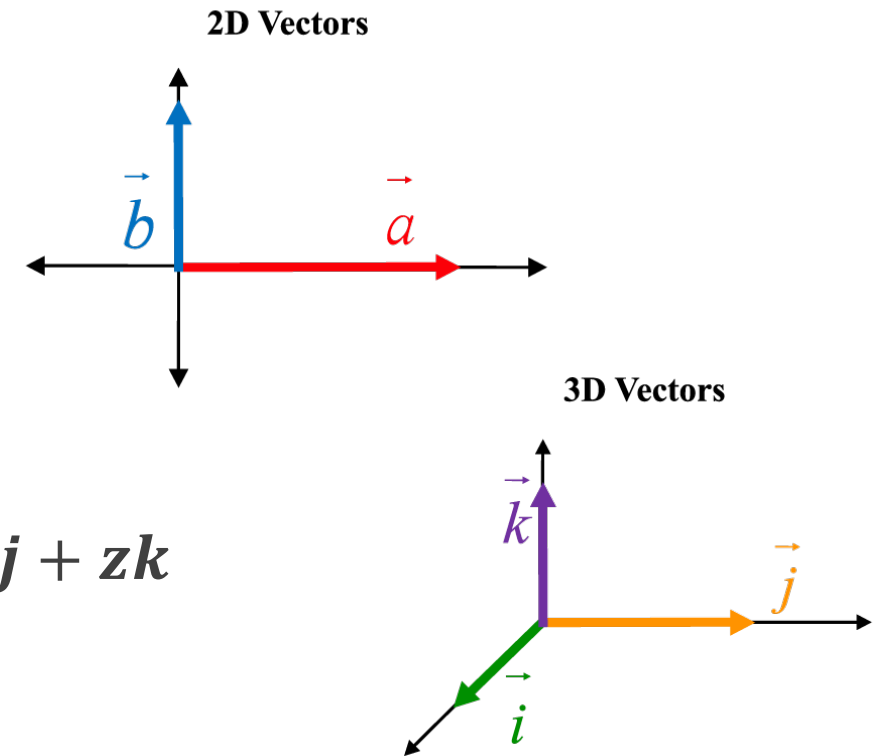
What are 3D vectors

- 3D vectors are just vectors with 3 values rather than 2
- They're used for real-world 3D spaces
- They can be used to visualise position as well as velocity
- They're used for analysing structures, fluid flows, stress, and movement in machines or vehicles.



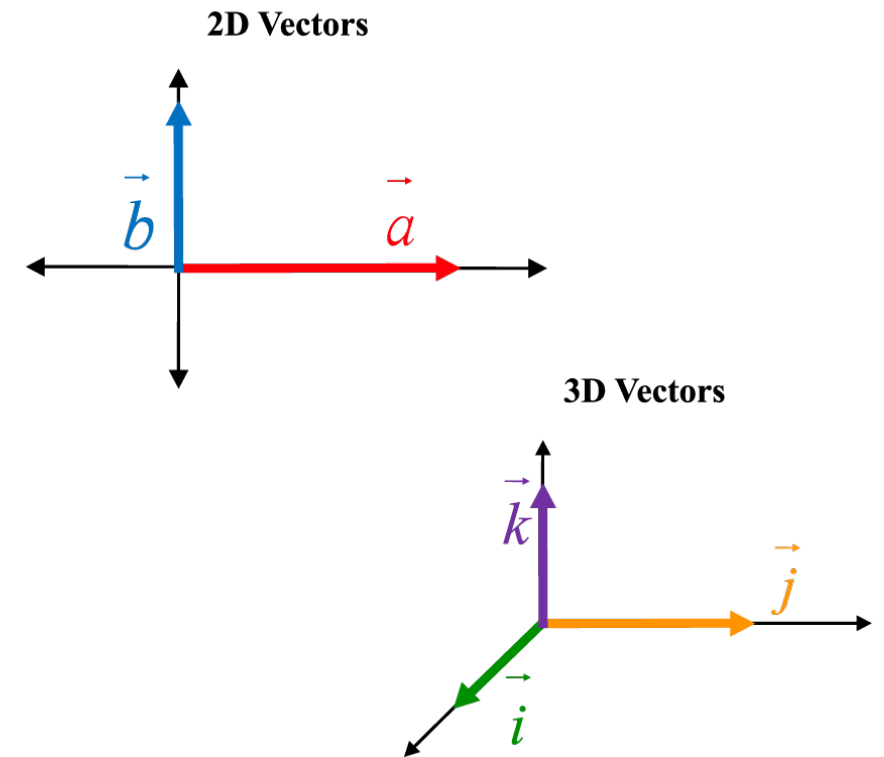
What are 3D vectors

- 2D vectors have x and y (a and b)
- 3D vectors have x, y and z (i, j, k)
- We can write 2D vectors as $\mathbf{v} = x\mathbf{i} + y\mathbf{j}$
- We can do the same with 3D vectors $\mathbf{v} = x\mathbf{i} + y\mathbf{j} + zk$



Example of 3D vector types

- Column = $\begin{pmatrix} 3 \\ -2 \\ 4 \end{pmatrix} = 3\mathbf{i} - 2\mathbf{j} + 4\mathbf{k}$



Calculating values with 3D vectors

- Both magnitude and resultant (vector addition) use the same method of calculation as 2D vectors

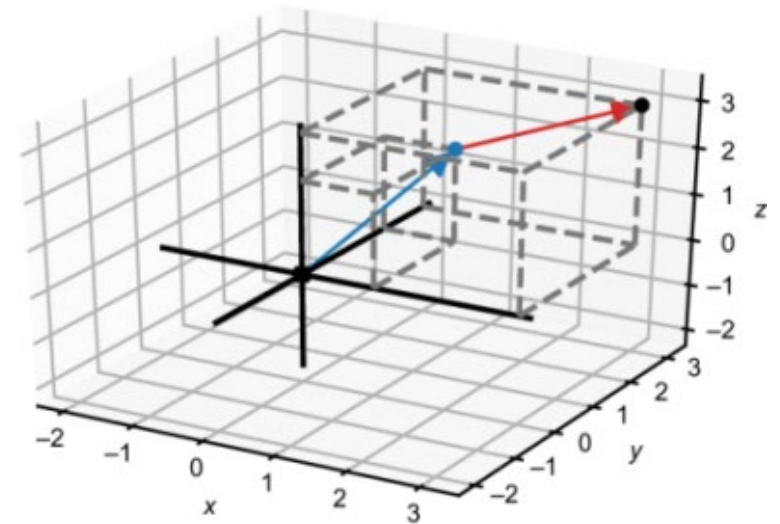
- Magnitude: $|a| = \sqrt{x^2 + y^2 + z^2}$

- Resultant: $\begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} + \begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix} = \begin{pmatrix} x_1 + x_2 \\ y_1 + y_2 \\ z_1 + z_2 \end{pmatrix}$

MAGNITUDE

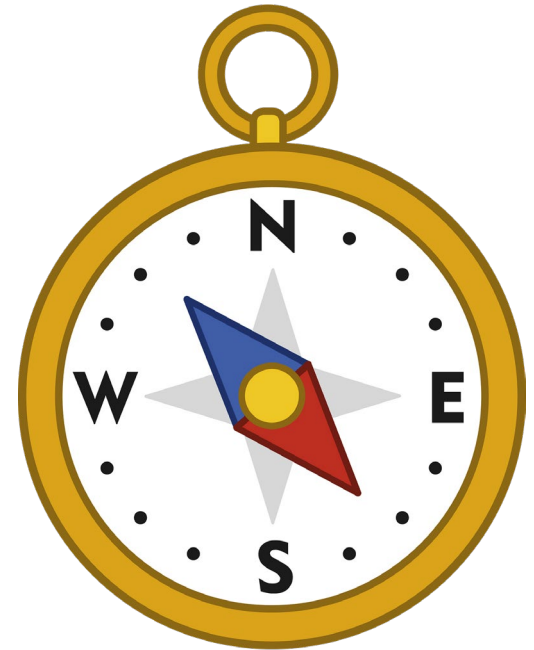
$$|a| = |x\mathbf{i} + y\mathbf{j} + z\mathbf{k}| = \left| \begin{pmatrix} x \\ y \\ z \end{pmatrix} \right| = \sqrt{x^2 + y^2 + z^2}$$

$$(1, 2, 2) + (2, 1, 1) = (3, 3, 3)$$



Normalising 3D Vectors

- We normalise 3D vectors to separate magnitude and direction, we do this by making magnitude 1
- This is helpful as it allows us to apply any magnitude to that vector
- Simple analogy:
 - Think of a compass needle.
 - The compass shows you direction (north, east, etc.).
 - It doesn't matter if the needle is long or short — the direction is what matters.
 - Normalized vectors are like compass needles: size = 1, direction = important.



Steps to normalise a 3D vector

- To find the normalised vector, we follow these steps:

- 1) Calculate the magnitude of the vector ($|V_1|$)
- 2) Put the magnitude and the vector into this equation, to give you a whole new vector.

$$U_1 = \left(\frac{x}{|V_1|}, \frac{y}{|V_1|}, \frac{z}{|V_1|} \right)$$

What to do with a normalised 3D Vector

- If you are given a magnitude in the question usually the question will then expect you to multiply the magnitude by the normalised vector
- So you'll get:

$$U_1 = N * \left(\frac{x}{|V_1|}, \frac{y}{|V_1|}, \frac{z}{|V_1|} \right)$$

Where N is the new magnitude, we are putting in

Example Question

- We have a force acting in direction $V_1 = \begin{pmatrix} 4 \\ 6 \\ -2 \end{pmatrix}$ with a magnitude of 12N we need to find the normalised vector then work out the vector for that magnitude
- Step 1:
Let's work out the magnitude of the vector ($|V_1|$)
$$|V_1| = \sqrt{4^2 + 6^2 + 2^2} = 7.483314774$$

Example Question

- Step 2:

Let's calculate the new normalised vector

$$U_1 = \left(\frac{4}{7.483}, \frac{6}{7.483}, \frac{-2}{7.483} \right) = (0.535, 0.802, -0.267)$$

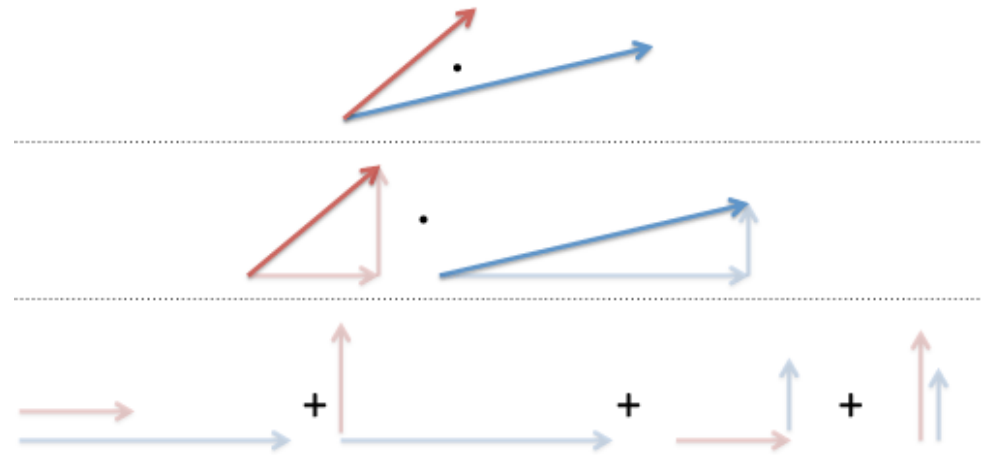
- Step 3:

Now let's add in our new magnitude

$$12 * (0.535, 0.802, -0.267) = (6.414, 9.621, -3.207)$$

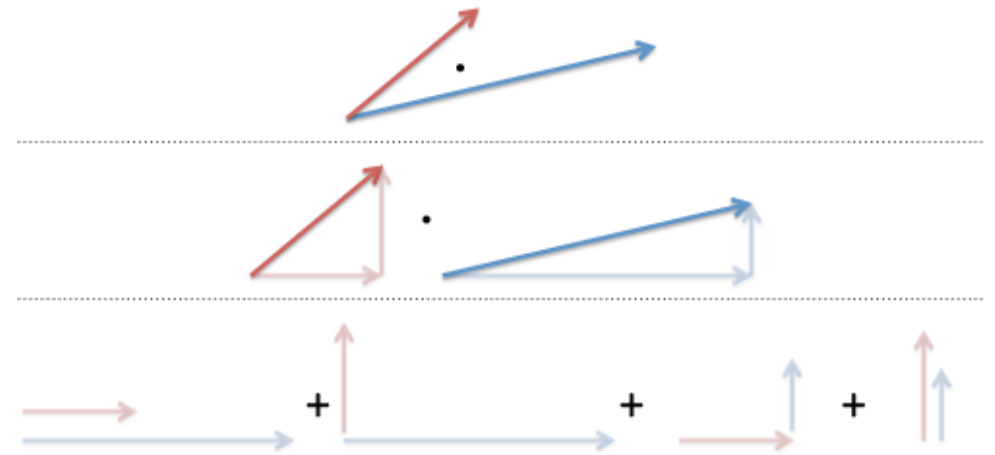
Dot Product

- Measures how much one vector acts in the direction of another
- Output is a **scalar** (number)
- Why it matters:
 - Work done: $W = \vec{F} \cdot \vec{D}$
 - Power flow in electrical systems
 - Checking if vectors are perpendicular (dot = 0)



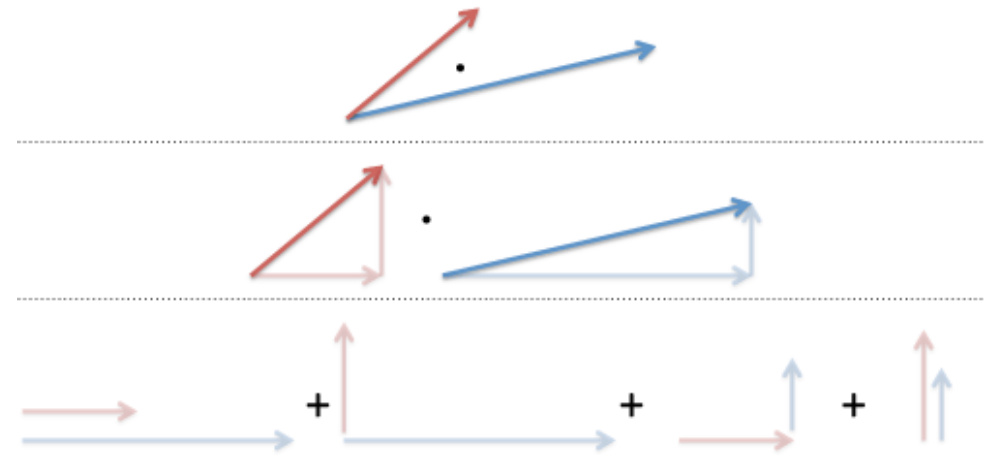
Dot Product

- There are two ways to calculate the dot product
 - $a \cdot b = |a||b| \cos \theta$
 - Or
 - $a \cdot b = (a_x * b_x) + (a_y * b_y)$
- It mainly depends on if you have cartesian or polar coordinates



Example using Dot Product (2D)

- Say we have two forces acting on a component $\mathbf{F1} = (3, 4)N$, $\mathbf{F2} = (5, 0)N$
 - To work out the dot product we just do:
 - $a \cdot b = (a_x * b_x) + (a_y * b_y)$
 - $F_1 \cdot F_2 = (3 * 5) + (4 * 0) = 15$



Example using Dot Product (3D)

- Two force vectors act on a joint $\mathbf{A} = (2, -1, 3)$, $\mathbf{B} = (4, 0, -2)$
 - To work out the dot product we just do:
 - $a \cdot b = (a_x * b_x) + (a_y * b_y) + (a_z * b_z)$
 - $A \cdot B = (2 * 4) + (-1 * 0) + (3 * -2)$
 - $A \cdot B = 8 + 0 - 6 = 2$

